

Wave function network

Exploring the 2D Ising Model through Wave Function Network (WFN)

We use classical Monte Carlo simulations to generate samples for the 2D Ising model, with each sample representing a snapshot of the system at a given temperature. These snapshots can be viewed as nodes within a wave function network. The connections between the nodes are determined by their Hamming distances. By analyzing the behavior of the distribution of these distances, we explore the structure of the wave function network across different temperatures and relate our understandings to the classical phase transition in the 2D Ising model.

Each wave function snapshot, labeled by an index j , takes the form:

$$X_j[w] = \{s_1^j, s_2^j, s_3^j, \dots, s_N^j\}, \quad (1)$$

where, s_n^j is the spin in j^{th} snapshot for n^{th} lattice site and w is an external parameter related to the snapshots, for example the kind of coupling or the temperature.

For the sake of simplicity, we also denote the set of all snapshots as:

$$X[w] = \{X_j\}_{N_r}, \quad (2)$$

N_r is the total number of snapshots that we have taken in MC simulations. Also, we assume no repetitions are present in a monte carlo simulations (this we do need to take care of, during noting down the snapshots from simulations).

1 Humming distances

To associate the notion of distances between the snapshot, we use humming distance,

$$d(X_i, X_j) = \sum_{p=1}^N |s_p^i - s_p^j| \quad (3)$$

We define a (geometric) network from our data sets by adopting the following procedure:

1. Each point X_i in the data set represents a node.
2. If two nodes are at distance $R > d$, we draw a link.
3. The distance R is chosen in a way that is dependent on the number of samples taken and reflects the typical value of distances for a given set of external parameters w . In particular, we define R as:

$$R = \frac{1}{N_r} \sum i = 1^{N_r} r_c(i), \quad (4)$$

where, r_c is the nearest neighbour distance of each node

The distribution for r_c , where c is the nearest neighbour humming distance for each node at different temperatures (from high temperature to critical temperature), is shown in 1

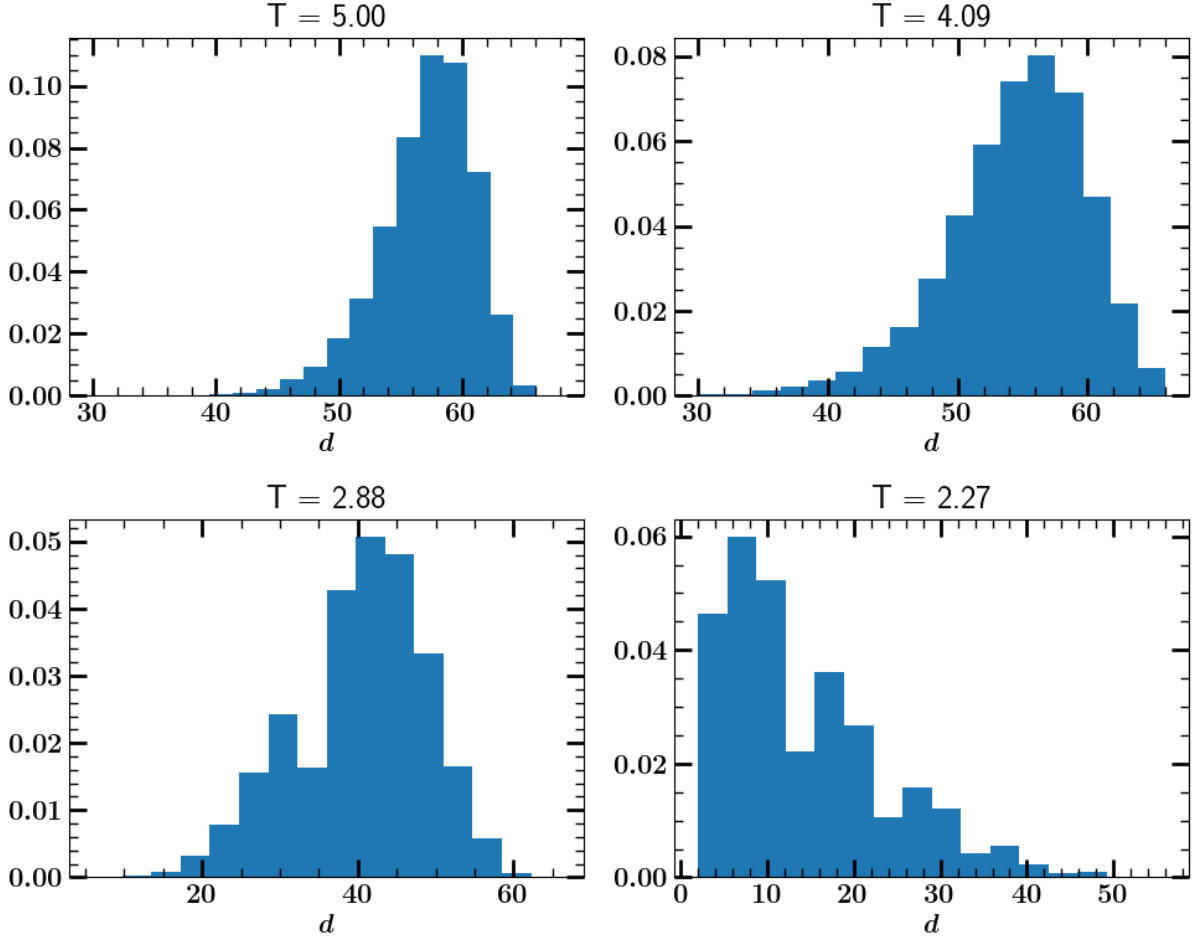


Figure 1: Histogram for the nearest neighbour distance for all the nodes in WFN.

The result in 1 is expected. As the temperature decreases and approaches the critical value, the peak of the nearest-neighbor distance distribution shifts to smaller values due to increasing correlation. At higher temperatures, where correlations are weaker, the mean distance is much larger, indicating weak correlation in the system.

2 Wave Function Network

The snapshots that we are generating are generated through some specific probability distribution, i.e., Boltzmann weight. Considering these snapshots as nodes and distances as correlations, can help us understand the behaviour of model through the network theory.

One can intuitively link physical properties to network characteristics. A wave function network with correlations will exhibit "hubs" — a few states with many connections and many states with few connections. In contrast, a random "infinite-temperature" state will primarily consist of states with an intermediate number of neighbours, lacking

both hubs and sparsely connected states.

The result for WFN at different temperatures are shown in figure 2, 3, 4 and 5. To understand the correlations in the wave function network (WFN) for the 2D Ising model, we analyze the degree distribution, which indicates how many connections (or edges) each node has with other nodes. At higher temperatures, as shown in figure 14, 13 and 9), the degree distribution reveals that only a few nodes are interconnected, signifying weak correlations. Near the critical temperature, the distribution broadens, showing a wide range of node connections, reflecting the enhanced correlations characteristic of criticality. The prominent peak for isolated nodes can be interpreted through percolation theory, where unconnected nodes correspond to disconnected clusters in the network.

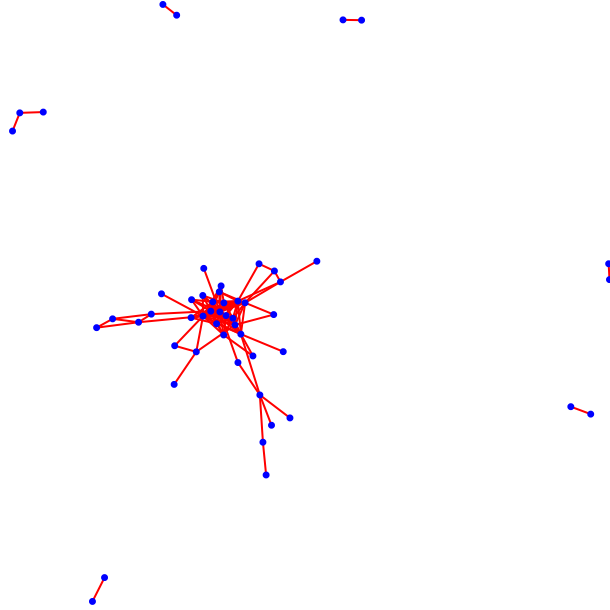


Figure 2: WFN at T_c for 100 samples. Blue points denotes the nodes, and red lines are the edges.

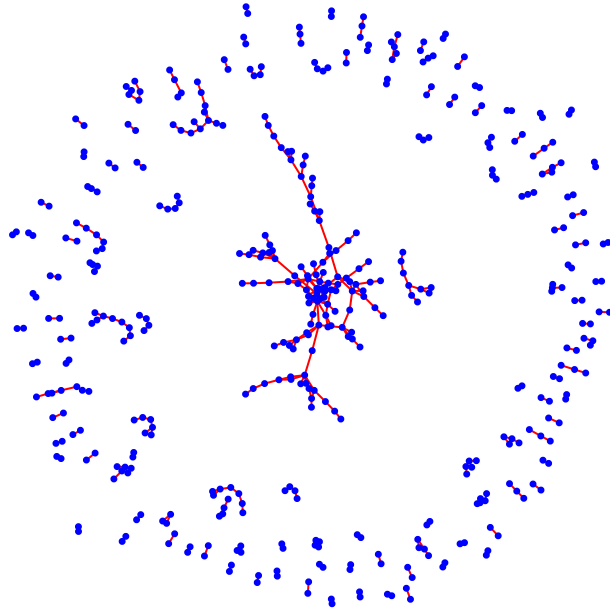


Figure 3: WFN at $T = 3.19$ for 1000 samples. Blue points denotes the nodes, and red lines are the edges.

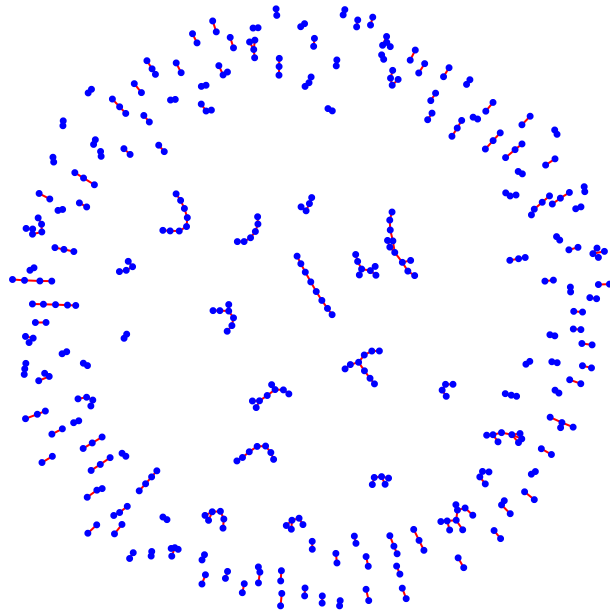


Figure 4: WFN at $T = 4.0$ for 1000 samples. Blue points denotes the nodes, and red lines are the edges.

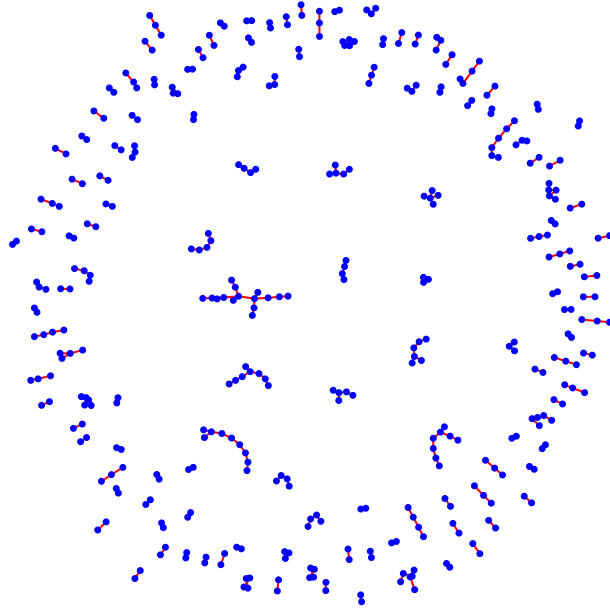


Figure 5: WFN at $T = 5.0$ for 1000 samples. Blue points denotes the nodes, and red lines are the edges.

Note: The networks shown here are generated using the Python library NetworkX, which is well-suited for creating networks based on bond-links (calculated through our analysis). In these graphs, the nodes with the most connections (links) are typically located at the center, while nodes with fewer connections are positioned farther from the center, following the spring layout of the NetworkX library.

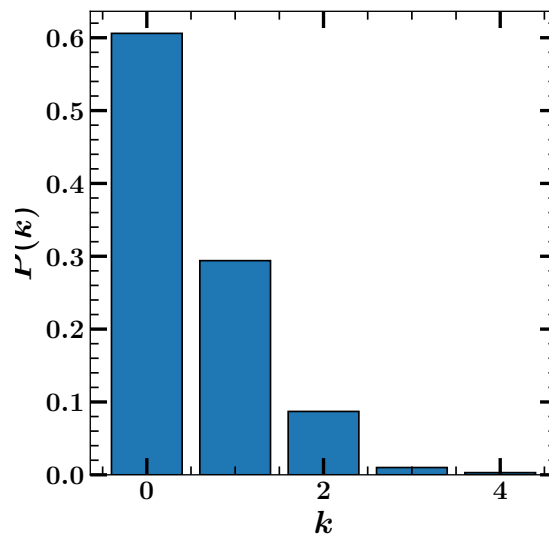


Figure 9: Degree distribution of the nodes (snapshots) at $T = 5.0$ (in units of $\frac{K_B T}{J}$) and 1000 samples).

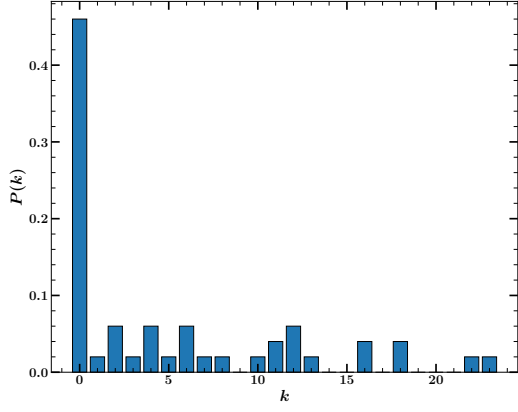


Figure 6: For 50 samples

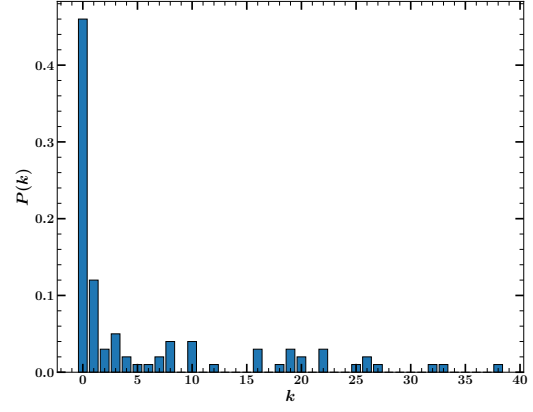


Figure 7: For 100 samples

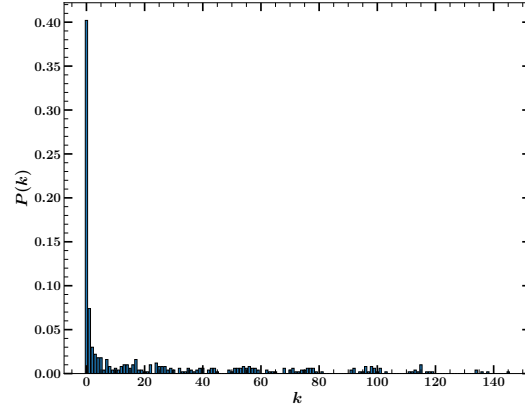


Figure 8: For 500 samples

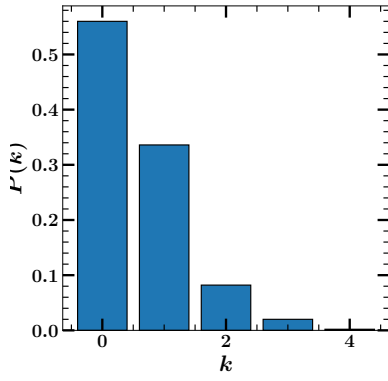


Figure 10: For 500 samples

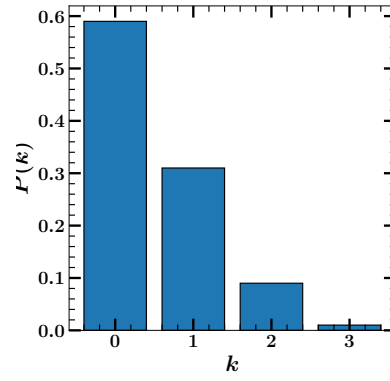


Figure 11: For 300 samples

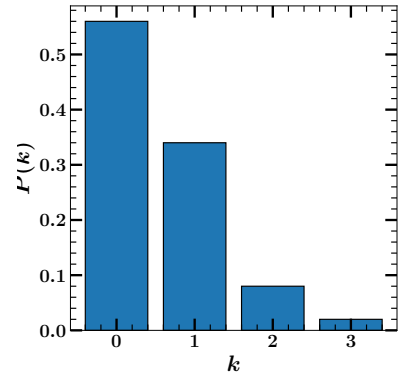


Figure 12: For 100 samples

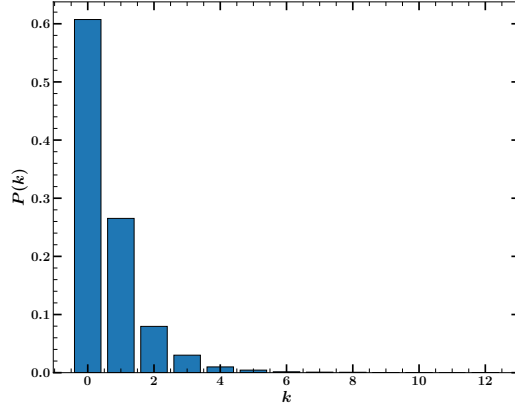


Figure 13: Degree distribution of the nodes (snapshots) at $T = 4.0$ (in units of $\frac{K_B T}{J}$) and 10000 samples).

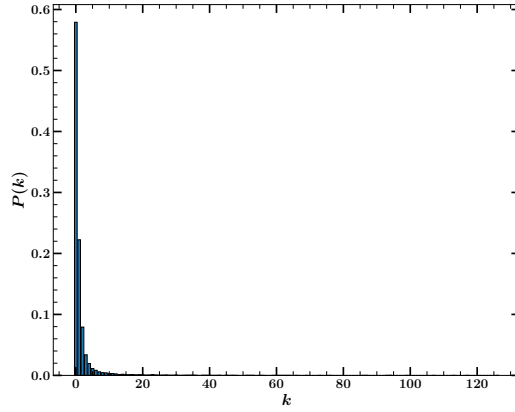


Figure 14: Degree distribution of the nodes (snapshots) at $T = 3.19$ (in units of $\frac{K_B T}{J}$) and 10000 samples).

Note: The networks shown here are generated using the Python library NetworkX, which is well-suited for creating networks based on bond-links (calculated through our analysis). In these graphs, the nodes with the most connections (links) are typically located at the center, while nodes with fewer connections are positioned farther from the center, following the spring layout of the NetworkX library.

3 Understanding the nature of such correlation

When we have understood that the signature of phase transitions are significant here, and this description of correlated states is reminiscent of what happens in several classes of scale-free networks: where the distribution becomes exponential for high temperature and it obeys the power law behaviour (scale in variant behaviour) at the critical temperatures.

When we fit the data to these predictions or just analyse there behaviour through log plots. We can also extract out the exponents from there, but the problem is to collect

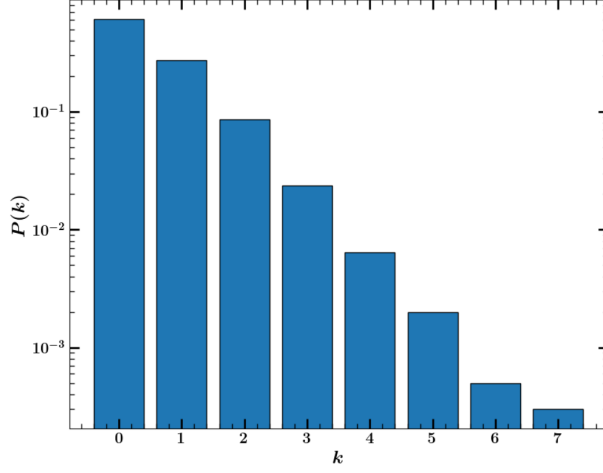


Figure 15: The degree distribution of the nodes (snapshots) at $T = 5.0$ (in units of $\frac{K_B T}{J}$ and 10000 samples), showing exponential behaviour.

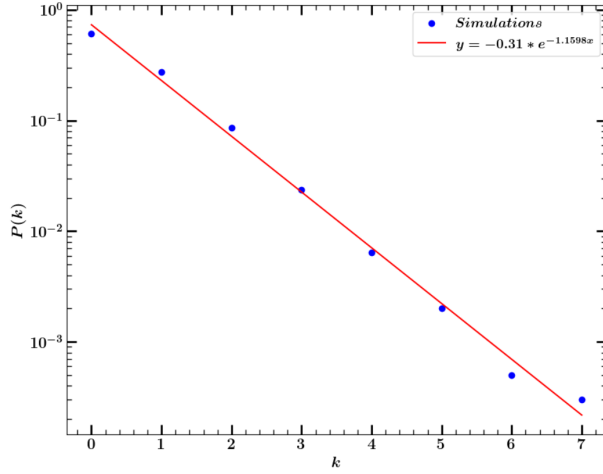


Figure 16: The fit for the degree distribution of the nodes (snapshots) at $T = 5.0$.

that much data (snapshots), and to simulate the system for large lattice sites, both of which are computationally heavy ways. The plots which might make some sense are shown in figure 17, 18, 15 and 16.

4 Q values

We also plot the distribution of Q-values, where Q is defined as the sum of the weights of a given node with all other nodes. The weights are inversely proportional to the Hamming distances between nodes.

At higher temperatures, where correlations are weak and the system behaves nearly randomly, the Hamming distances between snapshots are also nearly random. Consequently, the weights, being the reciprocals of these distances, lead to Q-values that are sums of nearly independent random variables. As a result, the distribution of Q-values follows a Gaussian or bell-shaped curve due to the Central Limit Theorem. This can be seen in figure 22.

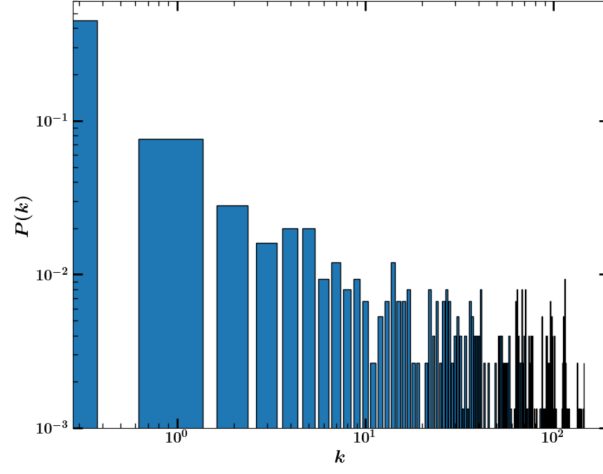


Figure 17: The degree distribution of the nodes (snapshots) at $T = 2.269$ (critical temperature). This was expected to show the power law behaviour.

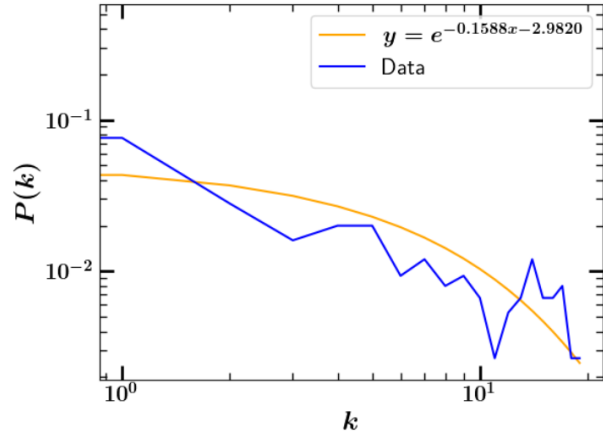


Figure 18: The fit for the degree distribution of the nodes (snapshots) at $T = 2.269$ (critical temperature).

In the critical regime, the system exhibits correlations across all length scales, resulting in a wide range of Hamming distances, from small to large. These distances are not purely random but are influenced by the underlying correlations. Therefore, the Q-value distribution reflects this complexity, lacking a well-defined peak and instead displaying a broader spread that mirrors the variety of correlations present. These plots are shown in figure 20, 21 and 19.

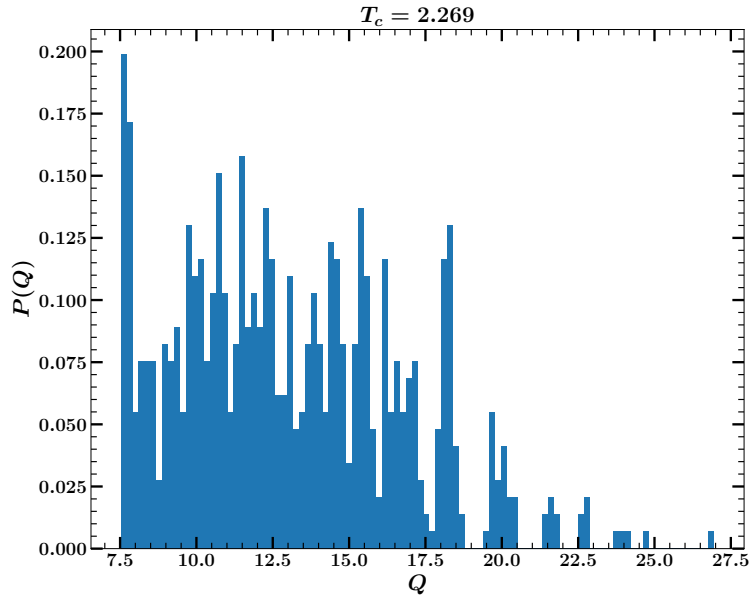


Figure 19: Distribution for the Q value at $T = T_c = 2.269$

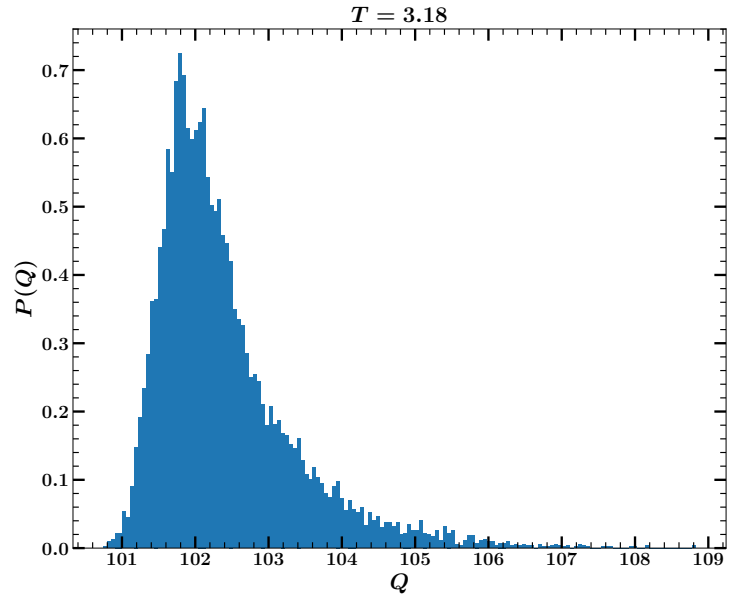


Figure 20: Distribution for the Q value at $T = 3.18$

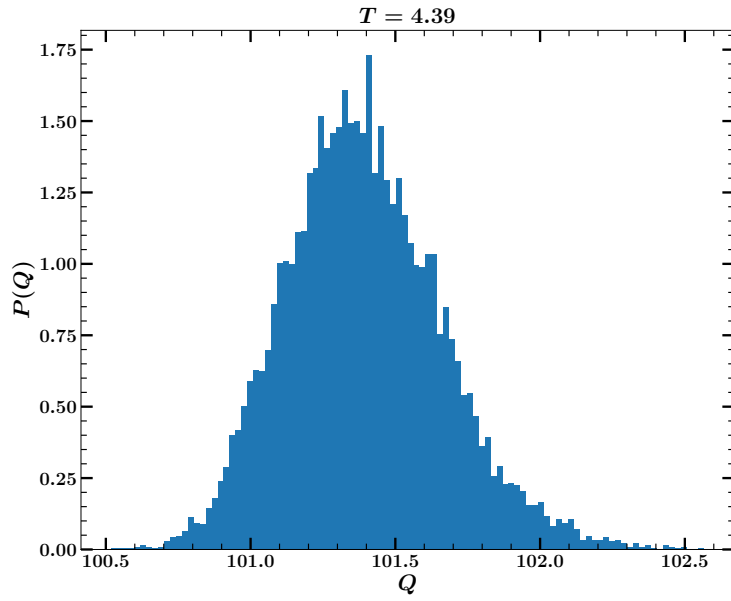


Figure 21: Distribution for the Q value at $T = 4.39$

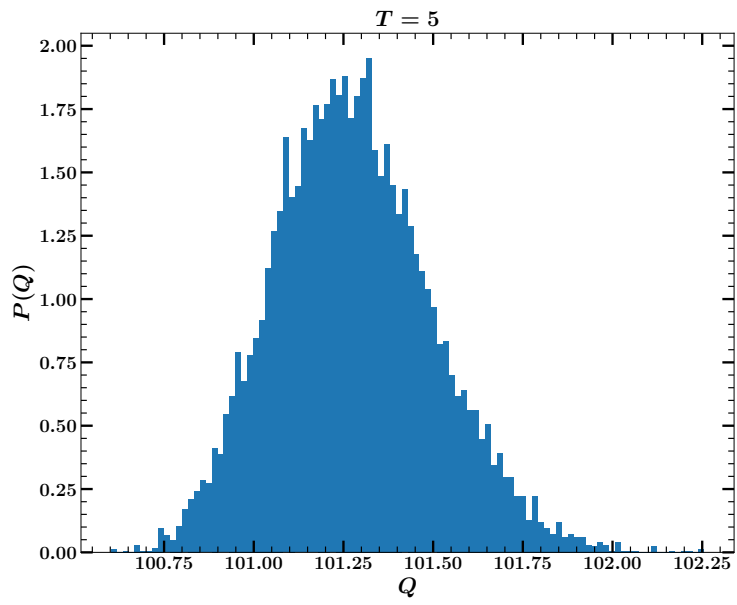


Figure 22: Distribution for the Q value at $T = 5$